

Lab Assignment -3

Course- B.Tech
Course Code- CSET343
Year- 4TH Year

Type- Specialization Elective
Course Name- AI in Healthcare
Semester- VII

Objective: Agenda: Perform data acquisition, cleaning, preprocessing and analysis on Tabular, Textual, Image, and Signal medical health data.

- Acquire and inspect real-world medical datasets across four different modalities.
- Apply modality-specific cleaning techniques, including healthcare-specific concerns like implausible clinical values, PHI (Protected Health Information) removal, and imaging metadata anonymization.
- Preprocess each data type into a model-ready format (scaled tensors, token sequences, image tensors, signal windows).
- For Tabular data, perform Hypothesis Test, Chi-square Test and ANOVA.
- Perform exploratory data analysis (EDA) appropriate to each modality.
- Apply Feature Engineering, Feature Selection & Extraction Noise Removal, Augmentation on relevant parts.
- Understand why healthcare data requires extra rigor compared to generic datasets (missingness encoded as valid-looking values, class imbalance, privacy constraints, signal noise).

Part	Data Type	Dataset	Link
A	Tabular	Pima Indians Diabetes Dataset	https://www.kaggle.com/datasets/jamaltariqcheema/pima-indians-diabetes-dataset
B	Textual	MTSamples Medical Transcriptions	MTSamples — Medical Transcriptions
C	Image	Chest X-Ray (Pneumonia) Dataset	Chest X-Ray Images (Pneumonia)
D	Signal	MIT-BIH Arrhythmia Database (ECG)	MIT-BIH Arrhythmia Database